

By examining the binary representation of numbers, you will understand why bits are numbered from 0 to 7, going from right to left. Bit 0 is " b_0 " and corresponds to 2^0 . Bit 1 is " b_1 " and corresponds to 2^1 , and so on.

Decimal	Binary	Decimal	Binary
0	00000000	32	00100000
1	00000001	33	00100001
2	00000010	.	
3	00000011	.	
4	00000100	.	
5	00000101	63	00111111
6	00000110	64	01000000
7	00000111	65	01000001
8	00001000	.	
9	00001001	.	
10	00001010	127	01111111
11	00001011	128	10000000
12	00001100	129	10000001
13	00001101		
14	00001110	.	
15	00001111	.	
16	00010000	.	
17	00010001		
.			
.			
.		254	11111110
31	00011111	255	11111111

Fig. 1.2: Decimal-Binary Table

The binary equivalents of the numbers from 0 to 255 are shown in Fig. 1-2.

Exercise 1.1: What is the decimal value of "11111100"?